Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations

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Abstract

Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable — yet investigations unfold under closure pressure that structurally favors premature convergence on a single explanation before alternatives have been meaningfully compared. I introduce Premature Accountability Convergence (PAC) as a concept that locates this failure not in individual cognition but in the interaction between closure pressure, documentation requirements, and the institutional incentives that make contained explanations more documentable than complete ones. I report a two-study empirical program with $N = 20$ GMP-experienced participants. In Study 1, a timed reflection protocol at three timepoints (0, +5, +10 minutes) establishes that comparison is time-sensitive: initial explanations were typically single-cause and person-centered, but reasoning broadened with additional time — suggesting that premature convergence reflects constrained investigative conditions rather than fixed reasoning limitation. Post-task discussion revealed that participants selected explanations oriented toward documentation acceptability rather than investigative completeness, choosing accounts that would satisfy audit review without implicating broader organizational conditions. In Study 2, an AI-supported reasoning scaffold

organized investigation into five stages enforcing explicit comparison before closure, with an AI component that detected convergence signals and triggered reopening prompts — functioning as a comparison enforcer rather than an explanation source. The scaffold was associated with broader causal accounts, delayed closure, and in several cases a shift from person-centered to systemic explanations. The most significant adoption barrier was organizational: participants questioned whether regulated organizations would deploy a tool that makes investigation reasoning more transparent, given that transparency surfaces alternatives that current practice leaves invisible. I introduce comparison preservation as a design principle for AI-supported reasoning systems and argue that PAC is as much an organizational problem as a cognitive one.

Keywords: premature accountability convergence, root cause analysis, comparison preservation, AI scaffolding, GMP, closure pressure, regulated environments, human-AI interaction

1 Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable. An investigation into a deviation should, in principle, compare multiple plausible explanations before arriving at a documented conclusion. In practice, investigations unfold under closure pressure — deadlines, documentation requirements, and procedural expectations that define when an investigation must be considered complete. Under these conditions, reasoning does not fail to generate alternatives; it fails to maintain them in a comparable state long enough for systematic evaluation. Convergence on a single explanation occurs not because other explanations have been eliminated, but because comparison between them has collapsed.

I conceptualize this dynamic as Premature Accountability Convergence (PAC): the early stabilization of causal accounts that occurs when the comparison between alternative explanations collapses before reasoning has fully developed. PAC is not treated here as a cognitive bias or individual reasoning failure. It is treated as a structural outcome —

a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure. This framing shifts the design question from "how do we help investigators think better?" to "how do we design workflows that keep comparison open longer?"

I report a two-study empirical program in pharmaceutical quality systems. In Study 1, I examine how reasoning develops under closure pressure in unassisted RCA tasks, using a timed reflection protocol with $N = 20$ GMP-experienced participants observed at three timepoints. Findings establish that early convergence is time-sensitive: initial explanations were typically single-cause, but reasoning broadened with additional time, suggesting that premature closure reflects constrained investigative conditions rather than fixed cognitive limitation. In Study 2, I introduce an AI-supported reasoning scaffold designed not to generate explanations but to preserve comparability between alternatives by delaying premature closure. The scaffold organizes reasoning into five stages — case intake, hypothesis expansion, PAC detection, structured comparison, and pre-closure verification — with an AI component that detects early convergence signals and triggers prompts that reopen the comparison space. Evaluated with $N = 20$ participants, the scaffold was associated with broader causal consideration, later closure, and in several cases a shift from person-centered to more systemic accounts.

The contributions of this work are threefold. First, I reframe premature convergence as a failure of comparison rather than a failure of idea generation, and introduce PAC as a concept that locates this failure in workflow structure rather than individual cognition. Second, I demonstrate empirically that comparison collapse is time-sensitive and therefore addressable through design — findings that provide an observational basis for scaffolded intervention. Third, I introduce comparison preservation as a design principle for AI-supported reasoning systems, and demonstrate one instantiation of this principle that maintains human interpretive authority while countering the structural conditions that produce premature convergence.

2 Related Work

Causal Explanation as Organizational Outcome

The foundational premise of this study — that causal explanations in regulated investigations are shaped by the organizational conditions under which they are produced, not only by the quality of individual reasoning — draws on Vaughan's (1996) account of the Challenger launch decision. Vaughan shows that the engineers and managers at NASA did not reach a catastrophic decision through negligence or bad faith; they reached it through a series of incremental, locally reasonable judgments that were structured by production pressure, communication patterns, and the normalization of prior deviations. Applied to GMP deviation investigations, this framework implies that the causal explanations produced by RCA reflect the conditions of deliberation, not only the evidence available. PAC names the specific mechanism through which this shaping occurs: comparison collapses, and the first viable explanation becomes the documented one.

Recognition-Primed Decision Making and the Satisficing Problem

Klein's (1998) account of recognition-primed decision making (RPD) provides the cognitive background against which PAC becomes legible as a design problem. Klein demonstrates that expert decision makers under time pressure do not compare options — they recognize a situation as familiar, generate a single candidate course of action, and evaluate it for workability before acting. This satisficing strategy is highly effective in dynamic environments where speed is essential. In regulated investigation contexts, however, the same expertise that enables rapid pattern recognition also enables rapid convergence — and the accountability stakes of investigation make that convergence consequential in a way that operational decision-making does not. The first viable causal explanation is adopted and elaborated without comparison to alternatives, not because alternatives are unavailable but because comparison is not built into the workflow. Expertise, in this context, accelerates the very dynamic that investigation quality requires resisting: the more experienced the investigator, the faster and more confident the initial convergence,

and the less likely the comparison space is to remain open. The problem PAC identifies is therefore not that investigators lack expertise, but that their expertise — channeled through RPD-style reasoning under closure pressure — produces premature convergence precisely when the accountability stakes of the explanation are highest.

Situated Action and the Conditions of Reasoning

Suchman's (1987) account of situated action establishes that reasoning is not plan-driven but shaped by the immediate conditions of its execution — what is visible, available, and interpretable at the moment a judgment must be made. An investigation workflow that requires a documented root cause before alternatives have been meaningfully compared creates conditions that structurally favor premature convergence, independent of individual reasoning quality. The scaffold developed in Study 2 is, in these terms, a redesign of the situated conditions of reasoning — an intervention in the environment rather than in the investigator.

Invisible Work and the Accountability Stakes of Comparison

Star & Strauss (1999) show that work performed in the gaps of formal systems tends to become invisible — not captured by indicators, not attributed to the conditions that produced it. In regulated investigation workflows, the comparative work of holding multiple causal explanations in parallel — evaluating their relative plausibility, identifying the evidence that distinguishes them, resisting premature commitment — is precisely this kind of invisible work. It is not required by the documentation template, not rewarded by closure expectations, and not visible in the final investigation record. The scaffold in this study makes this invisible comparative work structurally required, and therefore visible as a design concern.

Human-AI Interaction and Comparison-Preserving Design

The AI component of the Study 2 scaffold is designed around a principle that Amershi et al. (2019) formalize in their guidelines for human-AI interaction: AI systems should support

human control, make their capabilities and limitations clear, and avoid producing outputs that pre-empt or substitute for human judgment. In regulated investigation contexts, this principle takes on particular force: any system that generates causal conclusions or influences the direction of reasoning toward a specific explanation risks introducing a compliance risk. The AI component in this scaffold is therefore constrained to a specific and narrow function: detecting signals of premature convergence and triggering prompts that reopen the comparison space. It does not suggest causes, evaluate hypotheses, or generate conclusions.

Prototype-Based Evaluation

The methodological approach builds on established HCI practice for prototype-based evaluation. Nielsen (1994) demonstrates that think-aloud evaluation with small participant samples reliably surfaces the usability and reasoning support properties of an interface. Hutchinson et al. (2003) establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe: the technology probe makes visible the conditions it is designed to address. The present study applies this dual function to investigation reasoning: the scaffold is evaluated not only as an interface but as an instrument for surfacing the dynamics of comparison collapse and recovery under closure pressure.

Prior work has not examined premature causal convergence as a workflow design problem amenable to HCI intervention, nor has it introduced comparison preservation as a design principle for AI-supported reasoning in regulated investigation environments. This study addresses both gaps.

3 Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation investigations in these environments are formal quality events: they require documented root cause analysis, corrective and

preventive action planning, and approval by qualified reviewers before closure. These requirements impose closure pressure — temporal and procedural expectations that define when an investigation must be considered complete — that shapes how reasoning develops before alternatives have been fully compared.

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team. All participants had direct experience with GMP deviation investigation workflows, including root cause analysis, CAPA documentation, and quality review processes. Recruiting the same participants for both studies ensured consistent baseline familiarity with investigation conventions, but may have introduced shared response tendencies and inflated consistency across studies. This limitation is acknowledged in the scope section below.

Investigation Scenario

Both studies used the same standardized deviation scenario, constructed to reflect a realistic GMP quality event without reproducing proprietary case content. The scenario described an out-of-specification result in an analytical testing context, presented with sufficient contextual detail to support multiple plausible causal interpretations — including person-centered explanations (analyst error, training gap), procedural explanations (method ambiguity, SOP non-compliance), equipment-related explanations (instrument calibration, reagent preparation), and organizational explanations (workload pressure, oversight gap). The scenario was designed so that no single explanation was obviously correct, and so that premature convergence on a person-centered cause was a realistic and common response.

Study 1: Timed Reflection Protocol

Study 1 was designed to establish an empirical baseline for how reasoning develops under closure pressure without structured support. Participants were asked to produce a causal

explanation suitable for documentation within a fixed timeframe, using a standard RCA template. No additional support for structuring comparison was provided.

Reasoning was observed at three timepoints: an initial explanation produced immediately after reading the scenario (0 minutes), a revised or elaborated explanation after five minutes of unstructured reflection (+5 minutes), and a final explanation after ten minutes (+10 minutes). At each timepoint, participants recorded their current causal account and rated their confidence in it. The goal was to observe whether additional time changed the breadth, direction, or confidence of reasoning — and in particular, whether single-cause explanations at the initial timepoint were revised toward more systemic accounts with additional time.

Responses were coded for causal breadth (number of distinct causal categories represented), causal type (person-centered, procedural, equipment-related, organizational, or mixed), and evidence of comparison (explicit reference to alternative explanations or reasons for selecting one explanation over others).

Study 2: Comparison-Preserving Scaffold

Study 2 introduced an AI-supported reasoning scaffold designed to preserve comparison between alternative explanations by structuring the investigation workflow around explicit comparison steps before closure. The scaffold was implemented as a Streamlit web application and organized reasoning into five sequential stages:

1. **Case intake**: structured capture of deviation context without commitment to a causal conclusion

2. **Hypothesis expansion**: generation of multiple causal candidates across person, procedure, equipment, environment, and organizational categories

3. **PAC analysis**: rule-based detection of convergence-oriented language, with prompts to surface underexplored contributors and reopen alternative pathways

4. **Structured comparison**: explicit side-by-side evaluation of hypotheses before any single explanation is selected

5. **Pre-closure verification**: a required check confirming that alternatives were considered before final attribution

An AI component was integrated into Stage 3. When the system detected signals of premature convergence — single-cause fixation, convergence-oriented language, or movement toward closure before alternatives had been articulated — it triggered prompts requiring the participant to introduce additional alternatives or explicitly contrast competing explanations. The AI component did not suggest causes, evaluate hypotheses, or generate conclusions; it functioned exclusively as a conditional trigger for reopening the comparison space.

This design reflects a principled inversion of the dominant model for AI in RCA support. Suggestion engines, automated root cause generators, and pattern-matching classifiers treat AI as a source of explanations that investigators evaluate — a model that accelerates convergence by providing an anchoring explanation before the investigator has explored alternatives independently. The scaffold inverts this logic: AI functions as a comparison enforcer, acting against premature convergence rather than contributing to it.

Participants completed the same investigation scenario as Study 1 using the scaffold, and were asked to produce a final causal explanation at the end of the five-stage workflow. Post-task measures included ease of use, perceived reasoning support, causal breadth of the final explanation, evidence of comparison in the reasoning record, and adoption intent.

Scope and Limitations

Both studies are exploratory and pilot in scope. The use of the same participant pool across both studies limits the independence of findings and introduces potential carry-over effects. The single synthetic scenario may not generalize to the full range of deviation types encountered in live quality systems. Neither study establishes effects on downstream quality outcomes — investigation accuracy, CAPA effectiveness, or regulatory compliance.

4 Findings

Study 1: The Time-Sensitivity of Comparison

Convergence is Fast

At the initial timepoint, the large majority of participants produced single-cause explanations — typically person-centered accounts attributing the deviation to analyst error, training inadequacy, or procedural non-compliance. These explanations were produced quickly and with moderate to high confidence. With five additional minutes of unstructured reflection, alternative contributors began to appear: procedural ambiguities, equipment-related factors, and organizational conditions that had not been represented in the initial account. By ten minutes, participants either broadened their explanation or became more explicitly uncertain about the initial cause.

This pattern establishes a core finding: comparison is time-sensitive. The collapse of comparison that produces PAC is not a fixed property of how investigators reason; it is a property of how much time is available for reasoning to develop. When the investigation workflow defines closure before ten minutes of reflection have elapsed — as is commonly the case in practice — the reasoning state that gets documented is the one produced at the initial timepoint, not the one that would have emerged with additional time.

The Selection of Documentable Explanations

Post-task discussion revealed a finding more consequential than the timepoint data alone. Participants reflected that their real-world investigation practice rarely allows ten minutes of focused causal reasoning — and more importantly, that the explanation they select in practice is not simply the most plausible one but the most documentable one. Participants described selecting explanations that minimized the scope of the deviation — preferring individual-level causes over systemic ones — while maintaining logical defensibility. The criterion was not investigative completeness but documentation acceptability: an explanation that would satisfy audit review without opening lines of inquiry that might implicate broader organizational conditions.

This finding reframes PAC. Premature convergence in regulated investigations is not only a cognitive phenomenon produced by time pressure; it is also a rational strategy under institutional pressure. When investigators are accountable for the explanations they document — when a systemic account might trigger broader remediation requirements or invite scrutiny of organizational practices — selecting a contained explanation is not a reasoning failure. It is a structurally rational response to an accountability environment that rewards closure over completeness.

Study 2: Scaffold Effects

Causal Breadth and Delayed Closure

With the scaffold, most participants produced broader causal accounts than in Study 1. The five-stage structure required participants to generate multiple hypotheses before committing to any, and the PAC detection layer triggered reopening prompts in several sessions where convergence toward a single explanation had begun to occur. Participants maintained multiple hypotheses in parallel longer than in unassisted reasoning, and final explanations more frequently included procedural, organizational, or systemic contributors alongside person-centered causes.

In several cases, the scaffold produced a qualitative shift in the type of explanation: participants moved from accounts attributing the deviation to an individual actor to accounts that identified the procedural or organizational conditions that made the individual's action possible. This shift — from person-centered to systemic causation — is directly relevant to accountability conditions in regulated environments, where person-centered explanations tend to produce individual-level corrective actions while systemic explanations tend to produce structural remediation.

Usability and the Conceptual Barrier

Interface usability ratings were consistently positive. Participants did not struggle with the scaffold's five-stage structure, found the PAC detection prompts useful for identifying when they had been moving toward premature closure, and generally reported that the tool

kept alternatives visible longer than they would have maintained them independently. The adoption barrier was not usability but conceptual: the idea that investigation reasoning deserves structured support — that comparison should be treated as a required step rather than an optional one — was initially unfamiliar.

The Organizational Adoption Problem

The most significant finding from post-task discussion concerned the structural conditions of the tool's potential deployment. Participants raised a concern that went beyond interface design: whether GMP organizations would adopt a system that makes investigation reasoning more transparent. If a scaffold requires investigators to document that they considered systemic and organizational causes before settling on a person-centered explanation, the resulting record makes visible the alternatives that were available and not chosen. In organizations where investigation practice currently allows those alternatives to remain unconsidered, a transparency-enforcing tool changes not only how investigations are conducted but what they reveal about how they were conducted.

Participants did not express this as a criticism of the scaffold but as a realistic assessment of organizational incentive structures: regulated organizations operate under audit pressure that rewards closed investigations over thorough ones, and tools that make the investigation process more legible may surface accountability conditions that current practice obscures. This finding identifies the organizational precondition for adoption: a comparison-preserving scaffold cannot be deployed in isolation from a quality culture that values investigative completeness over documentation efficiency.

Summary

Together, the two studies establish that premature accountability convergence is both time-sensitive and institutionally rational. PAC is not primarily a problem of insufficient reasoning support; it is a problem of misaligned incentive structures that make premature convergence the rational response to institutional pressure. The scaffold demonstrates that structured support can delay convergence and broaden causal accounts. Whether that

broadening is organizationally sustainable depends on conditions that interface design alone cannot address.

5 Discussion

PAC as a Structural Problem

The central contribution of this study is a reframing: premature accountability convergence in regulated investigations is not primarily a cognitive problem amenable to reasoning support, but a structural problem produced by the interaction between closure pressure, documentation requirements, and the accountability stakes of causal explanation. The finding that participants selected explanations oriented toward documentation acceptability rather than investigative completeness — choosing contained, person-centered accounts that would satisfy audit review without opening systemic lines of inquiry — is not evidence of poor reasoning. It is evidence of rational adaptation to an institutional environment that rewards closed investigations over thorough ones.

This extends Vaughan's (1996) account of organizational decision-making. The present study shows the same dynamic operating at the level of individual investigation workflows: the explanation that gets documented is shaped not only by the evidence available but by the accountability environment in which the investigator operates. PAC names this dynamic specifically — the collapse of comparison that occurs when institutional pressure makes premature convergence the rational response.

AI as Comparison Enforcer

The design contribution of this study is the introduction of AI as a comparison enforcer rather than an explanation source. The dominant model for AI in investigation support treats AI as a producer of candidate explanations that investigators evaluate — a model that accelerates convergence by providing an anchoring explanation that narrows the comparison space. The scaffold inverts this logic entirely. The AI component's function is not to suggest what caused the deviation but to detect when comparison has collapsed

and to require its reopening. It acts against the structural tendency toward premature convergence rather than reinforcing it.

This instantiates the judgment-preservation principle that Amershi et al. (2019) formalize, but in a more specific and demanding form. In a high-accountability regulated context, preserving human judgment means designing AI to resist the institutional pressures that produce premature convergence — not merely to avoid automating the final decision. The AI in this scaffold preserves human judgment not by staying out of the way but by actively maintaining the conditions under which genuine judgment is possible.

The Organizational Adoption Problem

The most significant finding of this study is not about the scaffold's effectiveness but about the conditions of its deployment. Participants identified a structural barrier to adoption that interface design cannot address: regulated organizations may not want investigation reasoning to be more transparent, because transparency makes visible the alternatives that were available and not chosen. When an investigation record documents that systemic and organizational causes were considered before a person-centered explanation was selected, it also documents that those alternatives existed — and that the organization chose not to pursue them.

This finding connects the HCI concern of the present study to the broader theoretical question that motivates the PAC framework: how accountability is structured in regulated sociotechnical organizations. The scaffold demonstrates that comparison can be preserved through design. Whether that preservation is organizationally sustainable depends on whether the accountability environment rewards investigative completeness — a question that interface design alone cannot answer.

Limitations and Future Work

Several limitations bound the interpretation of these findings. The use of the same participant pool across both studies introduces potential carry-over effects and limits the independence of findings. The single synthetic scenario may not generalize to the

full range of deviation types and severity levels encountered in live quality systems. The organizational adoption concern raised by participants was not empirically investigated and would require separate study to examine systematically. The study does not establish effects on downstream quality outcomes, investigation accuracy, or regulatory compliance.

Future work should examine PAC in live investigation settings, where the institutional pressures that produce premature convergence are fully operative rather than simulated. The organizational adoption question is particularly important: understanding whether and under what conditions quality organizations would deploy comparison-preserving scaffolding — and what organizational changes would be required — is a research question that goes beyond HCI and into the organizational sociology of regulated work.

6 Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a structural account of why causal reasoning in regulated investigations stabilizes prematurely — not because investigators fail to generate alternatives, but because the comparison between alternatives collapses under institutional pressure before it has fully developed. Two empirical studies establish this claim and demonstrate one design response to it.

Study 1 showed that comparison is time-sensitive: initial explanations were single-cause and person-centered, but reasoning broadened with additional time. Post-task discussion revealed that premature convergence in practice reflects rational adaptation to an accountability environment that rewards documentable closure over investigative completeness. Study 2 demonstrated that an AI-supported comparison-preserving scaffold — organized around five stages of structured reasoning with an AI component functioning as a convergence detector rather than an explanation source — was associated with broader causal accounts, delayed closure, and in several cases a shift from person-centered to systemic explanation. The most significant barrier to adoption was organizational: participants assessed that regulated organizations might resist a tool that makes investigation reasoning more transparent, because transparency surfaces alternatives that current

practice leaves invisible.

Three contributions follow. First, PAC is introduced as a concept that locates premature convergence in workflow structure and institutional incentives rather than individual cognition — shifting the design question from reasoning support to comparison preservation. Second, AI as comparison enforcer is proposed as a design principle that inverts the dominant model of AI in investigation support, deploying computational mediation to resist convergence rather than accelerate it. Third, the organizational adoption problem is identified as a first-class research concern: the conditions under which comparison-preserving tools can be deployed in regulated environments are as much a question of organizational sociology as of interface design.

The most important direction for future work is examination of PAC in live investigation settings, where closure pressure, documentation requirements, and institutional incentives operate at full strength. The scaffold demonstrated here is a proof of concept; its translation into practice requires engagement with the organizational conditions that make premature accountability convergence structurally rational in the first place.

References

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (CHI '19). ACM.

Hutchinson, H., Mackay, W., Westerlund, B., Bederson, B. B., Druin, A., Plaisant, C., Beaudouin-Lafon, M., Conversy, S., Evans, H., Hansen, H., Roussel, N., & Eiderbäck, B. (2003). Technology probes: Inspiring design for and with families. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '03) (pp. 17–24). ACM.

Klein, G. A. (1998). *Sources of power: How people make decisions.* MIT Press.

Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.

Star, S. L., & Strauss, A. (1999). Layers of silence, arenas of voice: The ecology of visible and invisible work. *Computer Supported Cooperative Work, 8*(1–2), 9–30.

Suchman, L. A. (1987). *Plans and situated actions: The problem of human-machine communication*. Cambridge University Press.

Vaughan, D. (1996). *The Challenger launch decision: Risky technology, culture, and deviance at NASA*. University of Chicago Press.